Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

3.4 Internal Energy and Specific Heat Capacity

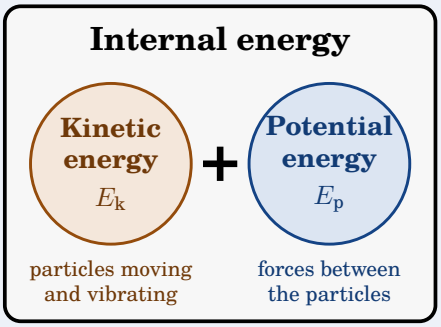
1 · Internal energy — the energy stored by the particles

Key concept — internal energy and temperature

Internal energy is the total kinetic and potential energy of all the particles (atoms and molecules) in a substance.

Temperature is a measure of the average kinetic energy of the particles.

	Temperature	E_k	E_p
(A) (C) (E)	rises	increases	constant
(B) (D)	constant	constant	increases



Review (Year 10): two equations to recall; neither is on the equation sheet.

$$E_k = \frac{1}{2} m v^2$$

- E_k: **kinetic energy** (J)
- m: **mass** (kg)
- v: **speed** (m/s)

$$E_p = m g h$$

- E_p: **gravitational potential energy** (J)
- m: **mass** (kg)
- g: **gravitational field strength** (N/kg)
- h: **height** (m)

Year 10 was **macro**: the energy of a whole object interacting with other objects, on a human-sized scale.

This unit is **micro**: the energy *inside* the object, on an atomic scale.

2 · Heating does one of two things — a hotter system, or a different state

Heating transfers energy *to* the particles, and the internal energy goes up in **one of two ways**. This sheet is the **left-hand branch**, Sections A, C and E of your Heating Curve; the right-hand branch, *latent heat* (Sections B and D), is the next sheet.

Heating — energy is transferred to the particles, so internal energy **increases**

The temperature rises
Heating Curve: Sections A, C, E
particles move and vibrate **faster**:
their **kinetic energy** goes up
e.g. water in a pan, 20 °C to 90 °C

The state changes
Heating Curve: Sections B, D
particles move **further apart**,
breaking free of their neighbours:
their **potential energy** goes up
e.g. ice at 0 °C melting to water at 0 °C: no temperature change

1) **Identify** for each situation whether the energy supplied is raising the **temperature** or changing the **state**, and name the particle energy store that increases (kinetic or potential). **[3]**

(a) A kettle warms water from 15 °C to 85 °C. *temperature rises — kinetic energy*

(b) Water in the kettle boils at 100 °C and turns to steam. *state changes — potential energy*

(c) A bar of chocolate softens and melts in a warm room. *state changes — potential energy*

3 · How much does the temperature rise? mass, material, energy in

Key concept — specific heat capacity

The temperature rise depends on the **mass**, the **material** and the **energy** put in.

Specific heat capacity, c, is the energy needed to raise the temperature of one kilogram of a substance by one degree Celsius.

$$\Delta E = m\,c\,\Delta\theta$$

- ΔE → **change in thermal energy** (J)
- m → **mass** (kg)
- c → **specific heat capacity** (J/kg °C)
- $\Delta\theta$ → **temperature change** (°C)

A C E

Specific heat capacity

Material	c / J/kg °C
Water	4200
Aluminium	900
Concrete	800
Iron	450
Copper	385
Lead	130

No need to memorise these. In an exam, the question gives you the value you need.

You will use them in calculations: use these values in every question on this sheet.

Worked example 1 — finding the energy

Calculate the energy needed to raise the temperature of 2.0 kg of water by 30 °C.

1. Write down $m = 2.0\,\text{kg}$ $c = 4200\,\text{J/kg °C}$ $\Delta\theta = 30\,\text{°C}$

2. Equation $\Delta E = m\,c\,\Delta\theta$

3. Substitute $\Delta E = 2.0 \times 4200 \times 30$

4. Answer $\Delta E = \mathbf{252\,000\ J}$ (that is 252 kJ)

Check: water at 4200 takes a few thousand joules per kilogram per degree, so 2 kg for 30 degrees is a few hundred thousand joules. The answer is the right size.

On every calculation, write the equation, the substitution and the answer with its unit.

2) **Calculate** the energy needed to raise the temperature of a 0.50 kg copper block by 15 °C. **[2]**

$\Delta E = m\,c\,\Delta\theta = 0.50 \times 385 \times 15 = \mathbf{2887.5\ J} \approx 2900\,J$

3) **Calculate** the energy needed to warm a 3.0 kg iron pan from 20 °C to 70 °C. Find the temperature *change* first. **[3]**

$\Delta\theta = 70 - 20 = 50\,\text{°C}$. $\Delta E = m\,c\,\Delta\theta = 3.0 \times 450 \times 50 = \mathbf{67\,500\ J}$

Worked example 2 — grams and kilojoules

A mug holds 250 g of water at 20 °C. Calculate the energy needed to heat it to 90 °C. Give your answer in kJ.

1. Convert to kg	$250\,\text{g} \div 1000 = 0.250\,\text{kg}$
2. Temperature change	$\Delta\theta = 90 - 20 = 70\,^\circ\text{C}$
3. Equation	$\Delta E = m\,c\,\Delta\theta$
4. Substitute	$\Delta E = 0.250 \times 4200 \times 70 = 73\,500\,\text{J}$
5. Convert to kJ	$73\,500\,\text{J} \div 1000 = \mathbf{73.5\ kJ}$

Check: a mass given in grams becomes a number *smaller than 1* in kilograms for anything you can lift in one hand. Joules to kilojoules is the same step: divide by 1000.

4) **Calculate** the energy needed to warm a 600 g aluminium saucepan by 40 °C. Give your answer in joules and in kilojoules. **[3]**

$m = 0.600\,kg$. $\Delta E = m\,c\,\Delta\theta = 0.600 \times 900 \times 40 = \mathbf{21\,600\ J = 21.6\ kJ}$

Worked example 3 — rearranging to find the temperature change

A heater transfers 42 000 J to 2.5 kg of water. Calculate the temperature rise.

1. Rearrange	divide both sides by $m\,c$: $\Delta\theta = \dfrac{\Delta E}{m\,c}$
2. Substitute	$\Delta\theta = \dfrac{42\,000}{2.5 \times 4200} = \dfrac{42\,000}{10\,500}$
3. Answer	$\Delta\theta = \mathbf{4.0\ °C}$

Check: put the answer back in the original equation: $2.5 \times 4200 \times 4.0 = 42\,000\,\text{J}$. It gives back the energy you started with.

5) **Calculate** the temperature rise when 18 000 J is transferred to a 0.80 kg iron block. **[3]**

$\Delta\theta = \dfrac{\Delta E}{m\,c} = \dfrac{18\,000}{0.80 \times 450} = \dfrac{18\,000}{360} = \mathbf{50\ °C}$

6) **Calculate** the specific heat capacity of a metal, given that a 2.0 kg block of it needs 7700 J to warm by 10 °C. Then use the table to identify the metal. **[4]**

$c = \dfrac{\Delta E}{m\,\Delta\theta} = \dfrac{7700}{2.0 \times 10} = \dfrac{7700}{20} = \mathbf{385\,J/kg\,°C}$. *The metal is* ***copper****.*

4 · Putting it to use — the same energy, two very different rises

7) **Calculate** the temperature rise of **(a)** 1.0 kg of copper and **(b)** 1.0 kg of water when each is given 8400 J of energy. **[4]**

(a) copper: $\Delta\theta = \dfrac{8400}{1.0 \times 385} = \mathbf{21.8\ °C}$ *(about 22 °C).* *(b) water:* $\Delta\theta = \dfrac{8400}{1.0 \times 4200} = \mathbf{2.0\ °C}$.

8) **Explain** why water is a good choice for the liquid pumped through central heating radiators and for a hot water bottle. Use your answers to question 9 and the idea of specific heat capacity. **[3]**

Water has a very high specific heat capacity (4200 J/kg °C), so each kilogram stores a large amount of energy for every degree it is heated. One kilogram of hot water therefore carries much more energy to a radiator, or to the bed, than a kilogram of most other materials at the same temperature, and because so much energy must leave it for its temperature to fall, it stays warm for a long time.

9) **Calculate** *(Higher)* the final temperature of 1.5 kg of water, starting at 20 °C, after an immersion heater has transferred 2000 J every second to it for 90 s. Give your answer to 2 significant figures. **[5]**

Energy transferred: $2000 \times 90 = 180\,000\,J$. $\Delta\theta = \dfrac{\Delta E}{m\,c} = \dfrac{180\,000}{1.5 \times 4200} = \dfrac{180\,000}{6300} = 28.6\,°\text{C}$. *Final temperature* $= 20 + 28.6 = 48.6 \approx \mathbf{49\ °C}$. *(1 mark energy, 1 rearrangement, 1 substitution, 1 temperature change, 1 final temperature to 2 s.f.)*